

**Gazette and Tender Management System along with an accessible website
for the Printing and Stationery Department, Government of Meghalaya**

*A Project Report submitted for partial fulfillment of the requirements for the final sem
of*

Masters of Computer Application (MCA)

of

JORHAT ENGINEERING COLLEGE

UNDER ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY



Submitted by:

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Final(4th) Semester

Jorhat Engineering College, Assam

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DEPARTMENT

OF

MASTER OF COMPUTER APPLICATION

JORHAT ENGINEERING COLLEGE

JORHAT, ASSAM



অসম চৰকাৰ
যোৰহাট অভিযান্ত্ৰিক মহাবিদ্যালয়
JORHAT ENGINEERING COLLEGE
DEPARTMENT OF M C A: JORHAT-07

Date:

CERTIFICATE

This is to certify that the project entitled “**Gazette and Tender Management System along with an accessible website for the Printing and Stationery Department, Government of Meghalaya**” submitted by Rijul Ali (Roll No: 230720043025) in partial fulfilment of requirements for the final **Semester** project of **Masters of Computer Application (MCA)** of **Jorhat Engineering College**, under **Assam Science and Technology University** has been examined.

Signature of the HOD

Dr. Siddhartha Baruah

Professor and HOD

Department of Master of Computer Application

Jorhat Engineering College, Jorhat



অসম চৰকাৰ
যোৰহাট অভিযান্ত্ৰিক মহাবিদ্যালয়
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Signature of the Internal Guide

Dr. Priyakshi Mahanta

Assistant Professor
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Jorhat Engineering College, Jorhat



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Signature of the external examiner

Name:.....

Designation.....

College/Institute.....

DECLARATION

I, **Rijul Ali**, Roll No. **230720043025**, an MCA 4th semester student of **Jorhat Engineering College**, do hereby declare that the project work entitled “**Gazette and Tender Management System along with an accessible website for the Printing and Stationery Department, Government of Meghalaya**”, a web-based application developed to streamline the uploading, publishing, and management of official gazettes and tender documents, is an original and authentic piece of work carried out by me at the **National Informatics Centre (NIC), Meghalaya State Centre, Shillong** under the guidance of **Mrs. Rosemary Maisnam** (Scientist-D & Joint Director, IT). for the partial fulfilment and award of the degree of Master of Computer Application. This project has not been submitted anywhere else for the award of any other degree/Diploma. Where other sources of information have been used, they have been acknowledged.

Rijul Ali

Roll No: 230720043025

Reg No: 002607223

Department of Computer Application

Jorhat Engineering College

Assam Science & Technology University

ACKNOWLEDGEMENT

I am deeply honored to have had the opportunity to pursue my **Master of Computer Application** under **Assam Science and Technology University**. This academic journey has provided me with a platform to explore the field of **web application development** and apply it in a real-world project — the **Gazette and Tender Management System** — within a short but enriching timeframe.

I would like to express my heartfelt gratitude to everyone who contributed directly or indirectly to the successful completion of this project. This acknowledgement is not a mere formality but a sincere expression of appreciation for the guidance, support, and encouragement I received throughout the process.

I extend my deepest gratitude to my external guide, **Mrs. Rosemary Maisnam** (Scientist-D & Joint Director, IT), **National Informatics Centre, Meghalaya State Centre**, for their invaluable guidance, motivation, and technical support. Their keen interest in the project and insightful suggestions played a crucial role in shaping the system.

I also take this opportunity to thank **Dr. Priyakshi Mahanta**, Assistant Professor, **Department of Computer Applications**, for their constant support, valuable feedback, and encouragement, which helped me stay focused and confident in completing this work successfully.

A special thanks to all my **classmates and batchmates**, who stood by me with helpful suggestions and moral support. I am equally thankful to my **respected faculty members at Jorhat Engineering College** for their academic guidance and encouragement throughout the MCA program.

Lastly, I am truly grateful to the **Almighty** for giving me the strength, patience, and clarity to carry this work to completion. Thank you all.

Rijul Ali

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ABSTRACT

The “**Gazette and Tender Management System**” is a web-based solution developed for the **Printing and Stationery Department, Government of Meghalaya**, aimed at streamlining the publication, management, and distribution of official gazette notifications and government tenders. Recognizing the need for modernization and transparency in handling official documents, this system provides a secure and efficient platform for content creators, publishers, and administrators to manage the end-to-end process of gazette uploads and tender publications.

The application features a role-based dashboard system, enabling authorized users to log in, upload PDF files, manage records, and monitor publication history. It ensures proper version control of gazette files and offers facilities for replacing or updating documents when necessary. Additionally, users can view and access published gazettes and tenders through an intuitive and searchable interface.

From an administrative perspective, the system integrates a robust backend infrastructure that manages user authentication, file storage, and metadata tracking. It ensures data integrity, traceability, and accessibility while maintaining the confidentiality and legal sanctity of official government documents.

In summary, the **Gazette and Tender Management System** enhances operational efficiency, promotes paperless governance, and contributes to the digitization efforts of the Meghalaya government. By offering a centralized platform for managing gazettes and tenders, the system facilitates better coordination between departments, improves document accessibility for citizens, and upholds transparency and accountability in public communication.

INTRODUCTION

The “**Gazette and Tender Management System**” is a web-based portal developed for the Printing and Stationery Department, Government of Meghalaya, with the goal of modernizing and streamlining the processes involved in managing, publishing, and accessing official government gazettes and tenders. This project leverages modern technology to make the uploading, publication, and retrieval of official documents more efficient, transparent, and user-friendly for both departmental staff and the general public.

The system provides different dashboards based on user roles such as **Creator, Publisher, and Admin**. Creators can securely upload gazette documents, specifying details such as the gazette part and file information. Publishers have the authority to review, approve, or request changes to uploaded gazettes before they are published for public access. Admin users manage user accounts, oversee system configurations, and ensure the integrity and security of the platform.

For public users, the system offers a convenient interface to search, view, and download gazettes and tender documents. This eliminates the dependency on physical copies and manual distribution, ensuring that official government publications are readily accessible and promoting greater transparency in government operations. The platform is built using **Spring Boot, Thymeleaf, Bootstrap, jQuery, and PostgreSQL**, providing a robust, scalable, and secure architecture. These modern technologies ensure that the system can handle increasing document volumes, deliver fast performance, and provide an intuitive user experience.

Overall, the **Gazette and Tender Management System** represents a significant step towards e-governance for the Government of Meghalaya. By digitizing the management and dissemination of government publications, the system enhances operational efficiency, reduces paperwork, and ensures that official information reaches citizens and stakeholders in a timely and accessible manner.

AIMS AND OBJECTIVES

Aim of the project:

The aim of the “**Gazette and Tender Management System**” project is to provide a centralized, efficient, and user-friendly online platform for the Printing and Stationery Department, Government of Meghalaya. The system is designed to streamline the processes of uploading, managing, publishing, and accessing government gazette notifications and tender documents. It seeks to improve operational efficiency, enhance transparency, and make official documents easily accessible to both departmental users and the general public while ensuring security and compliance with government standards.

Objectives of the project:

1. **Develop a User-Friendly Platform:** Build an intuitive, responsive platform to enable easy uploading, publishing, and searching of gazette and tender documents by department users and public stakeholders.
2. **Implement Role-Based Access Control:** Ensure secure system access by defining specific roles (Creator, Publisher, Admin) with distinct permissions and responsibilities.
3. **Facilitate Efficient Document Management:** Allow Creators to upload gazette files with associated details, enable Publishers to review and approve documents, and empower Admins to manage users, configurations, and system operations.
4. **Provide Public Access to Information:** Enable citizens to search for, view, and download gazette notifications and tender documents directly from the portal, fostering transparency and ease of access.
5. **Ensure Secure Transactions and Data Integrity:** Implement security measures like Spring Security to protect data during upload, storage, and retrieval processes, ensuring compliance with government data protection policies.
6. **Support Scalability and Future Expansion:** Utilize modern technologies such as Spring Boot, Thymeleaf, and PostgreSQL to build a system that is robust, scalable, and maintainable for future enhancements or increased document volumes.
7. **Enhance Administrative Efficiency:** Provide dashboards for administrators to monitor system usage, document uploads, and user activities to aid in better decision-making.
8. **Promote Environmental Sustainability:** Reduce reliance on physical printing and distribution of government gazettes by digitizing the publication process, contributing to eco-friendly practices.

FEASIBILITY STUDY

It is essential to evaluate the feasibility of a project at the earliest possible stage to ensure successful implementation and sustainable operation. The following feasibility studies were conducted to assess the viability of the **Gazette and Tender Management System**.

Operational Feasibility:

The proposed Gazette and Tender Management System is designed to be highly user-friendly, ensuring that departmental staff and public users can navigate it with ease. The system provides clear workflows for different roles, such as Creators, Publishers, and Admins, minimizing complexity and reducing the learning curve. It allows seamless document uploads, approvals, and searches, ensuring efficient day-to-day operations for the Printing and Stationery Department and convenient access for citizens.

Technical Feasibility:

The Gazette and Tender Management System leverages modern, robust technologies like Spring Boot, Thymeleaf and PostgreSQL. These tools ensure a secure, scalable, and high-performing platform capable of handling large volumes of document data. The department has the necessary hardware and network infrastructure in place to support deployment and ongoing operations. Technical evaluation confirms that the system can be effectively implemented using existing resources and skillsets.

Economic Feasibility:

A cost-benefit analysis indicates that the development and implementation costs of the Gazette and Tender Management System are justified by the significant benefits it offers. These include savings from reduced printing and distribution costs, time efficiency for staff, improved document management, and enhanced transparency. The shift to digital processes leads to long-term operational savings, making the financial investment highly worthwhile for the department.

Scheduling Feasibility

A realistic project timeline has been prepared for the development, testing, and deployment of the Gazette and Tender Management System. This schedule accounts for resource availability, technical complexities, and potential risks or delays. By identifying critical dependencies and maintaining careful planning, the project is expected to be delivered within the stipulated timeframe, ensuring timely operational readiness.

Legal and Regulatory Considerations:

The Gazette and Tender Management System adheres to relevant legal and regulatory standards, including data protection laws to ensure the privacy and security of sensitive government documents and user information. It respects government protocols regarding the publication and dissemination of official gazette notifications and tenders, ensuring compliance and maintaining the credibility of the department.

Conclusion:

The feasibility study confirms that the **Gazette and Tender Management System** for the Printing and Stationery Department, Government of Meghalaya, is a practical, beneficial, and sustainable initiative. The system is technically robust, economically viable, operationally efficient, and legally compliant, making it a valuable investment for enhancing document management and public accessibility to official government publications.

REQUIREMENT GATHERING AND ANALYSIS

System requirements are documented in the Software Requirement Specification (SRS), which serves as the official reference for what is expected from system developers. This document defines *what* the system should do, without specifying *how* it should be implemented. It ensures that the requirements are complete, consistent, and well-understood.

The Software Requirement Specification document fulfills the following purposes:

- Specifies the external system behavior.
- Defines constraints on the implementation.
- Is easy to modify as requirements evolve.
- Serves as a reference for system maintainers.
- Records careful considerations regarding the system's life cycle.
- Defines acceptable responses to undesired events or errors.

1. HARDWARE AND SOFTWARE SPECIFICATIONS

1.1 HARDWARE REQUIREMENTS

The Gazette and Tender Management System will run on any modern computer or mobile device with the following minimum hardware specifications:

- **Processor:** Dual-core Intel or equivalent (1.8 GHz or higher).
- **RAM:** 2GB or above.
- **Internal Storage:** 50 GB free disk space or above.
- **Display:** Modern display supporting at least 1366 × 768 resolution.
- **Network:** Reliable internet connection for accessing the web portal.

1.2 SOFTWARE REQUIREMENTS

The system will operate on client devices equipped with internet connectivity and a modern, JavaScript-enabled web browser. Preferred browsers include:

- Google Chrome (latest version recommended)
- Mozilla Firefox
- Microsoft Edge

Server-Side Technologies:

- Operating System: Windows or Linux
- Backend Framework: Spring Boot
- Frontend Technologies: Thymeleaf, HTML, CSS, JavaScript, Bootstrap
- Database: PostgreSQL
- Development Environment: Visual Studio Code, IntelliJ IDEA, or equivalent IDE
- Web Server: Apache Tomcat embedded in Spring Boot

SYSTEM STUDY

System study involves an in-depth examination of the various operations performed by the existing system and their interrelations both within and outside the system's boundaries. It serves as a structured approach to refine goals and establish clear objectives. The primary purpose of system development lies at the core of this process. One of the most effective approaches to system analysis is structured analysis.

The system study for the **Gazette and Tender Management System for the Printing and Stationery Department, Government of Meghalaya** encompasses evaluating the existing processes, identifying challenges, and proposing modern digital solutions to enhance operational efficiency, data security, and user experience. The following sections describe the essential components of this study:

1. Current System Analysis

The current management of government gazette notifications and tender publications is largely manual and paper-based. This results in several challenges:

- **Manual Processes:** Uploading, approving, and managing gazette notifications and tenders involve extensive paperwork and physical file movements between departments and officials.
- **Paper-Based Records:** Records related to gazette notifications, tender notices are maintained manually, increasing the risk of loss, damage, or misplacement of critical documents.
- **Limited Accessibility:** Stakeholders—including government officials, publishers, and the public—face delays in accessing published gazettes or tenders, as physical copies must be distributed or collected in person.
- **Inefficient Tracking:** Tracking the status of submitted gazettes or tenders, approvals, and publications is time-consuming and prone to errors due to the lack of a centralized digital system.

2. Proposed System Overview

The proposed Gazette and Tender Management System seeks to address these limitations by introducing a comprehensive digital platform that streamlines all processes related to gazette and tender management. Key features of the proposed system include:

- **Digital Upload and Management:** Authorized users can upload gazette documents and tender notices through a secure online portal, eliminating the need for physical submissions.
- **Role-Based Access Control:** Distinct dashboards and permissions for Creators, Publishers, and Admins ensure controlled access and security.
- **Approval Workflow:** Digital workflows enable efficient approval and verification of documents by authorized personnel, reducing turnaround times.
- **Centralized Digital Repository:** All gazette notifications and tender notices are stored electronically, allowing instant retrieval and search capabilities for both staff and the public.
- **Public Access:** Citizens can easily view and download published gazettes and tender notices from the department's website.
- **Audit and Tracking:** Every login within the system is logged for audit purposes, improving accountability and transparency.

3. Benefits of the Proposed System

The new Gazette and Tender Management System offers significant advantages:

- **Efficiency:** Streamlines document submission, approval, and publication processes, reducing manual effort and administrative delays.
- **Accessibility:** Provides online access to gazette notifications and tenders for all stakeholders, improving transparency and public awareness.
- **Accuracy:** Reduces errors associated with manual record-keeping and ensures precise tracking of submissions and approvals.
- **Security:** Incorporates robust security measures to protect sensitive government documents and ensure only authorized access to the system.
- **Environmental Impact:** Minimizes paper usage, supporting eco-friendly initiatives and promoting sustainable governance practices.

SYSTEM IMPLEMENTATION

System Overview

The **Gazette and Tender Management System** is a web-based application designed to streamline the management, approval, and publication of government gazettes and tender notices for the Printing and Stationery Department, Government of Meghalaya. The system ensures secure user authentication, efficient document workflows, and accessible publication for both government officials and the public.

Key Features:

- 1. User Authentication and Role-Based Access Control**
 - Separate dashboards for Creators, Publishers, and Admins
 - Secure login and session management
- 2. Digital Upload and Management**
 - Upload gazette documents and tender notices
 - Track document statuses (draft, submitted, approved, published)
- 3. Approval workflows**
 - Publishers review and approve documents digitally
- 4. Centralized Repository and Search**
 - Digital storage and retrieval of gazettes and tenders
 - Public access to published documents
- 5. Audit Trail and Logging**
 - Logs of secure login actions for security and accountability

Tools/Technologies and Languages Used for Development:

- **Frontend Development:** Thymeleaf, HTML, CSS, JavaScript
- **Backend Development:** Spring Boot
- **Database Management:** PostgreSQL
- **API Development:** RESTful APIs
- **Version Control:** GitHub

Implementation Plan

- 1. Frontend Development (Thymeleaf)**
 - Design user-friendly web pages for login, dashboards, document uploads, and viewing published gazettes and tenders
 - Implement responsive design to ensure seamless usability on desktops, tablets, and mobile devices
 - Provide validation feedback for user inputs to improve data integrity
- 2. Backend Development (Spring Boot)**
 - Develop RESTful APIs for:
 - User authentication and session management
 - Gazette and tender upload, approval, and retrieval processes
 - Role-based content access
 - Implement Spring Security for robust authentication and authorization mechanisms
 - Handle data persistence and retrieval using Spring Data JPA integrated with PostgreSQL
 - Implement exception handling to manage and report system errors effectively
- 3. Database Management (PostgreSQL)**
 - Design a normalized relational database schema to store:
 - User credentials and roles
 - Gazette tender and others document metadata
 - Approval logs and audit records
 - Optimize database queries for fast access and scalability
- 4. Version Control (GitHub)**

- Utilize GitHub for:
 - Code versioning and collaboration
 - Tracking changes and maintaining project history
 - Managing feature branches for efficient teamwork
- 5. **Integrated Development Environment (IDE):**
 - Use **IntelliJ IDEA** for backend development
 - Use **Visual Studio Code (VS Code)** for frontend development
- 6. **Deployment and Testing**
 - Deploy the web application on a secure server, ensuring public access to published gazettes and tenders
 - Conduct thorough testing:
 - Unit Testing for individual code modules
 - Integration Testing to validate data flow between components
 - User Acceptance Testing (UAT) with real users from the department

By following this structured implementation plan, the **Gazette and Tender Management System** will deliver an efficient, secure, and modern platform for managing government gazettes and tenders. It will empower the Printing and Stationery Department with improved workflow efficiency, greater transparency, and enhanced accessibility for both government stakeholders and the public.

MODULE DESCRIPTION

1. User Authentication Module:

- **Description:** The User Authentication Module serves as the foundation for secure access control within the Gazette and Tender Management System. It ensures that only authorized users can access specific functionalities based on their roles (Admin, Creator, Publisher).
- **Components:**
 - **User Login:** Provides users with a secure interface to authenticate themselves using their credentials (username and password). User roles determine the dashboard and features accessible after login.
 - **Role-Based Access Control:** Manages permissions for different user types, ensuring that only authorized users can perform sensitive operations like gazette publishing, tender uploads, or user management.
 - **Session Management:** Handles secure session tracking to maintain user authentication status during system use.

2. Gazette and Tender Management Module

Description: The Gazette and Tender Management Module manages the full lifecycle of gazette publications and tender notices within the department. Instead of treating gazette and tender as separate workflows, the system is structured around user roles—Creators and Publishers—who collaborate to prepare, review, and publish departmental documents.

Components:

Creator Module

- Document Creation and Upload:
 - Creators can create new gazette notifications or tender notices.
 - Upload PDF documents and enter related metadata (e.g., gazette part, tender number, title, publication date, submission deadlines, etc.)
- Draft Management:
 - Creators can save gazette and tender entries as drafts for further editing or review before submission.
 - Drafts can be edited or deleted as needed until finalized for submission.
- Submission to Publisher:
 - Once ready, Creators submit gazette or tender documents to the Publisher for approval and publishing.
- Submission History:
 - The Creator dashboard includes a history page that shows all submitted gazettes and tenders along with their current status (e.g., created, Submitted, Published).

Publisher Module

- Document Review and Approval:
 - Publishers review gazette and tender documents submitted by Creators.
 - Publishers can:
 - Approve and publish the document to the public website.
 - Send documents back to Creators for corrections or additional information.
 - Edit or delete documents if necessary.
- Publishing Controls:
 - Ensure that only reviewed and approved gazettes and tenders are visible to the public.
- Published History:
 - The Publisher dashboard includes a history page displaying all gazette and tender documents they have published, providing a record of publication activities for tracking and audit purposes.

3. **Administrative Dashboard Module:**

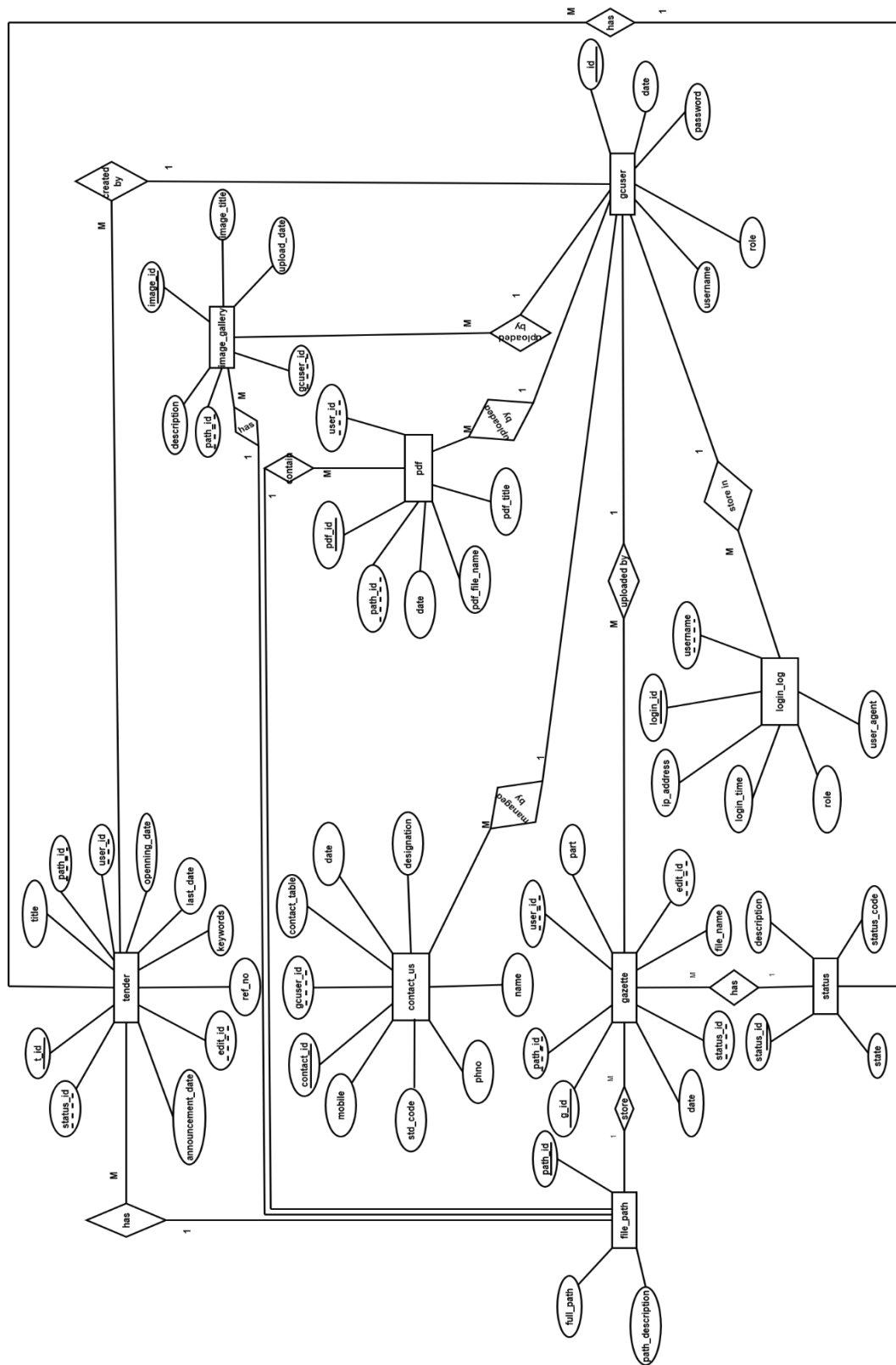
- **Description:** This module equips administrative users with comprehensive tools to oversee and manage system operations, ensuring smooth functionality of gazette and tender processes as well as dynamic content management for the public-facing website.
- **Components:**
 - **Gazette and Tender Monitoring:** Allows administrators to review and track gazette and tender records after they are published, ensuring accuracy and compliance before public availability.
 - **User Management:** Enables administrators to add, edit, or delete users in the Creator and Publisher roles, maintaining proper access controls and organizational structure.
 - **Notification Management:** Provides functionality for uploading and managing important notifications in PDF format, ensuring timely information dissemination to end users.
 - **Photo Gallery Management:** Allows admins to upload and curate images for the website's index page, keeping the public portal visually engaging and up to date.
 - **Contact Us Management:** Facilitates the addition and updating of contact information displayed on the public website's Contact Us page, ensuring accurate communication channels for users.

4. **Public Access Module:**

- **Description:** This module ensures that citizens and stakeholders can seamlessly access gazette and tender information published by the department.
- **Components:**
 - **Search Functionality:** Enables users to search for gazette notifications and tender documents using filters like department, date, gazette part, or keywords.
 - **Download Feature:** Allows users to view and download gazette documents and tender notices directly from the website.
 - **PDF Notifications View and Download:** Provides users with the ability to view and download important PDF notifications uploaded by the department for public information.
 - **Informational Pages Access:** Offers users access to general information pages such as *Contact Us*, *About Us*, and other sections maintained by the Printing and Stationery Department for public awareness and assistance.
 - **Responsive Design:** Ensures that the public portal is accessible and user-friendly across various devices, including desktops, tablets, and smartphones.

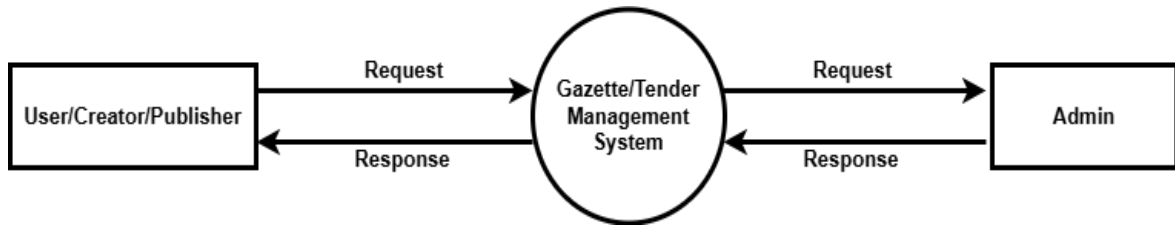
These modules collectively establish a comprehensive system that streamlines document management for gazettes and tenders while maintaining security, accountability, and ease of public access. The Gazette and Tender Management System ensures the Printing and Stationery Department can operate efficiently, transparently, and in alignment with modern e-governance initiatives.

ENTITY RELATIONSHIP DIAGRAM(ERD)

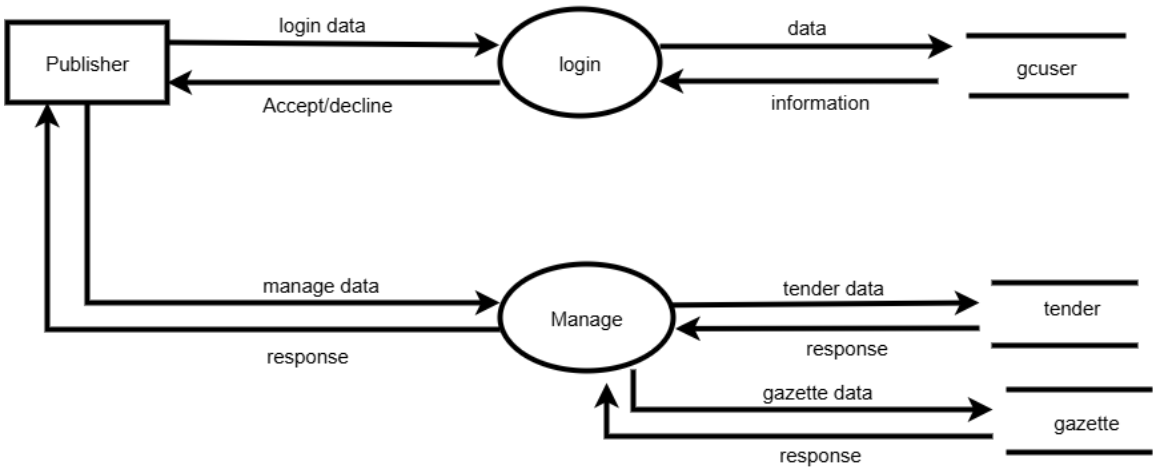
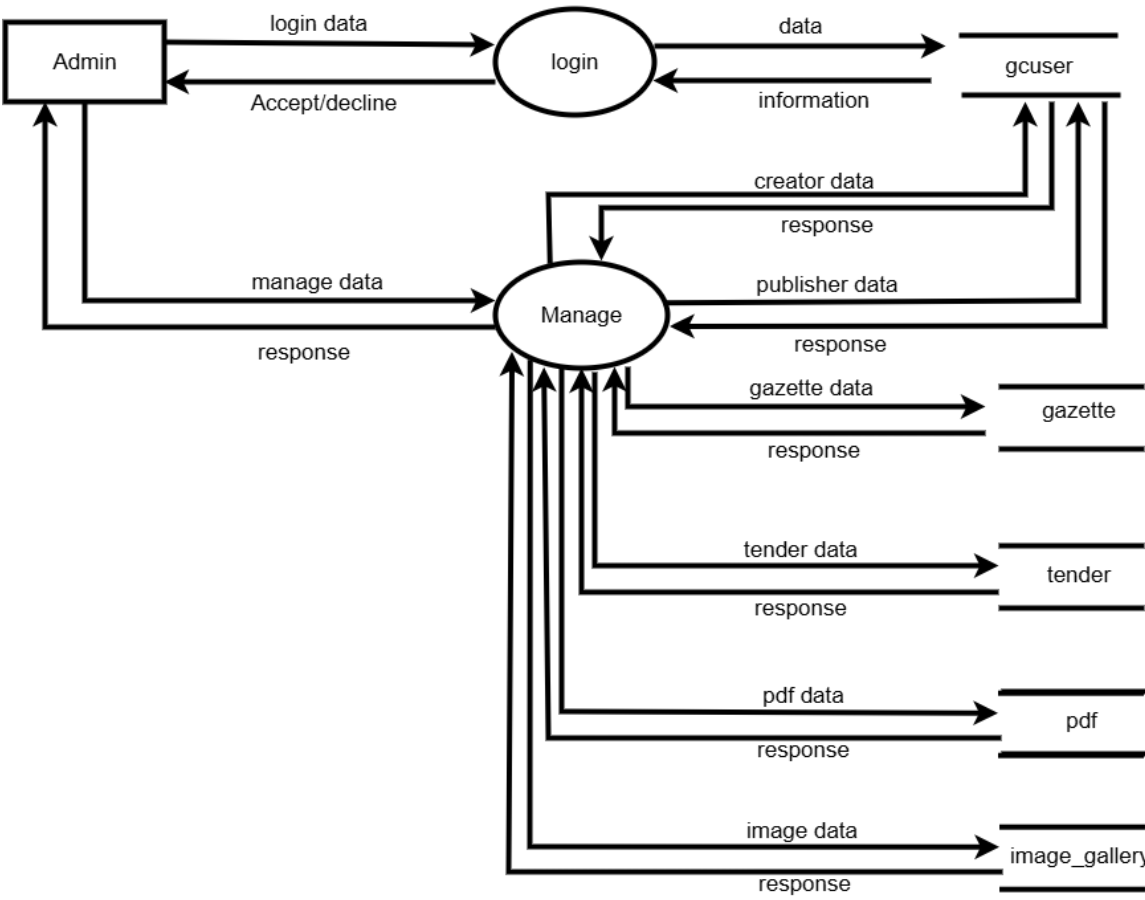


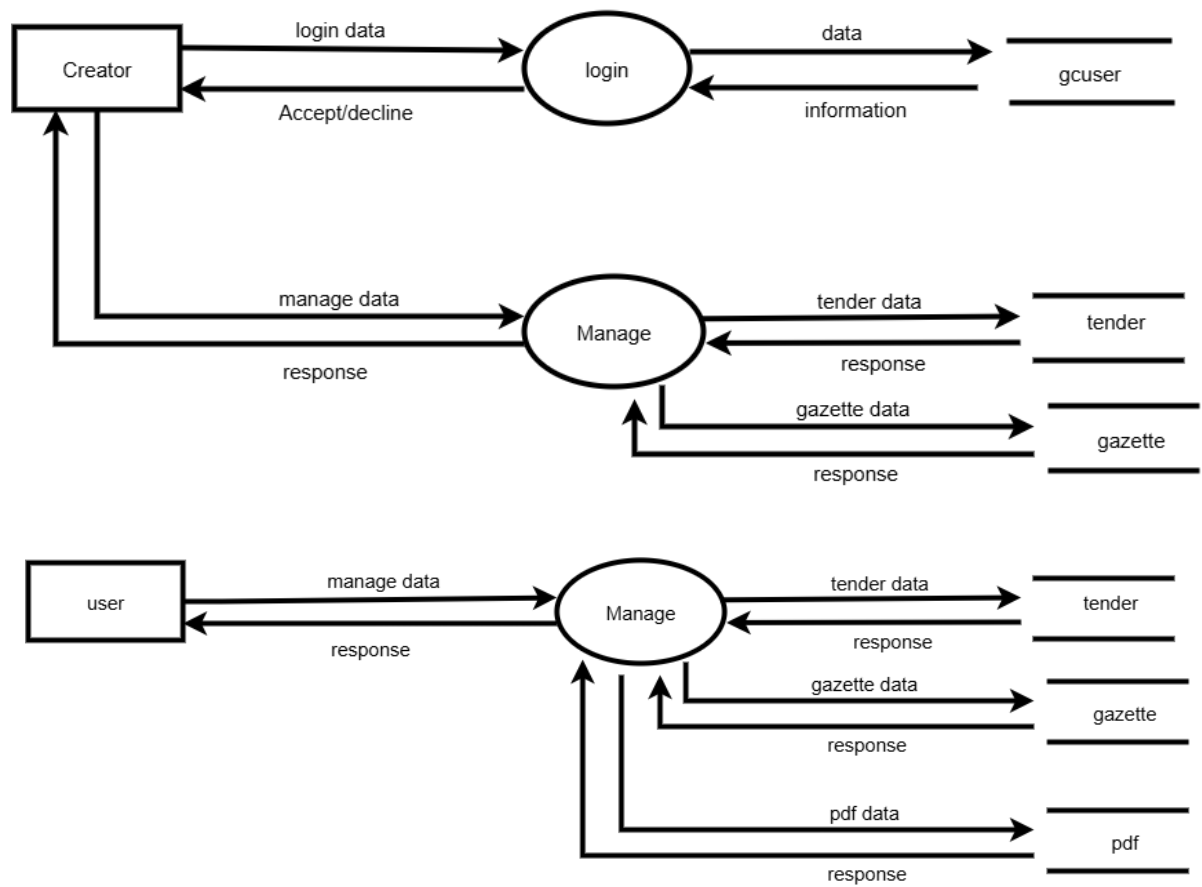
DATA FLOW DIAGRAM(DFD)

Level 0



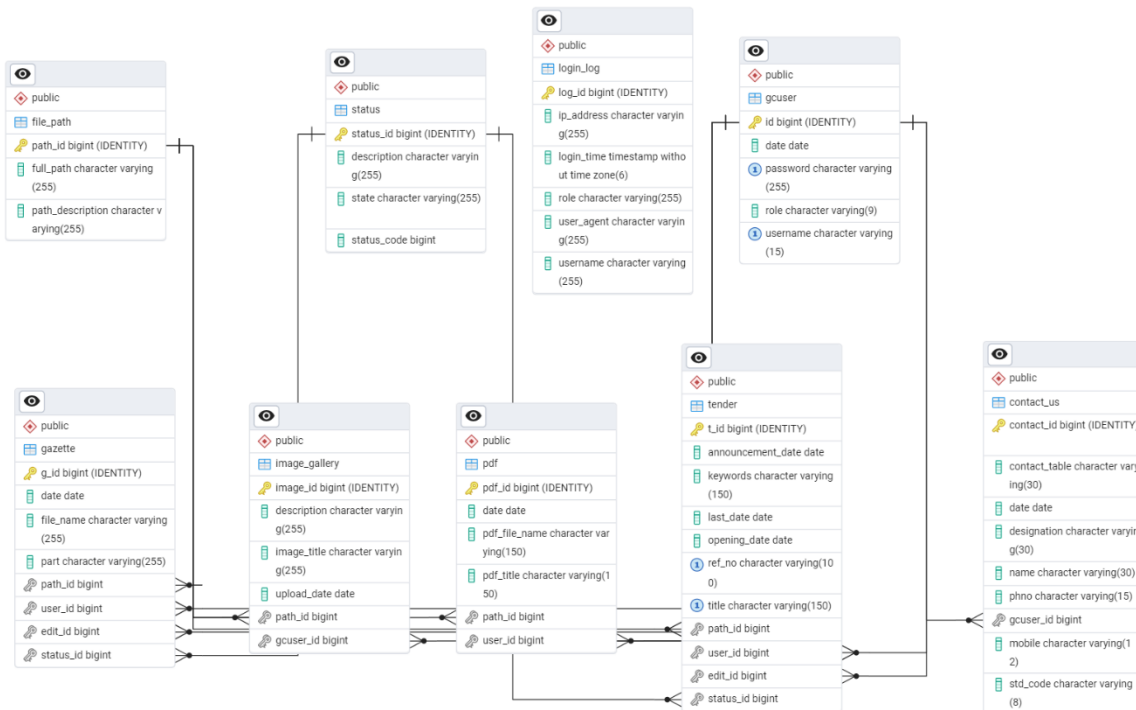
Level 1





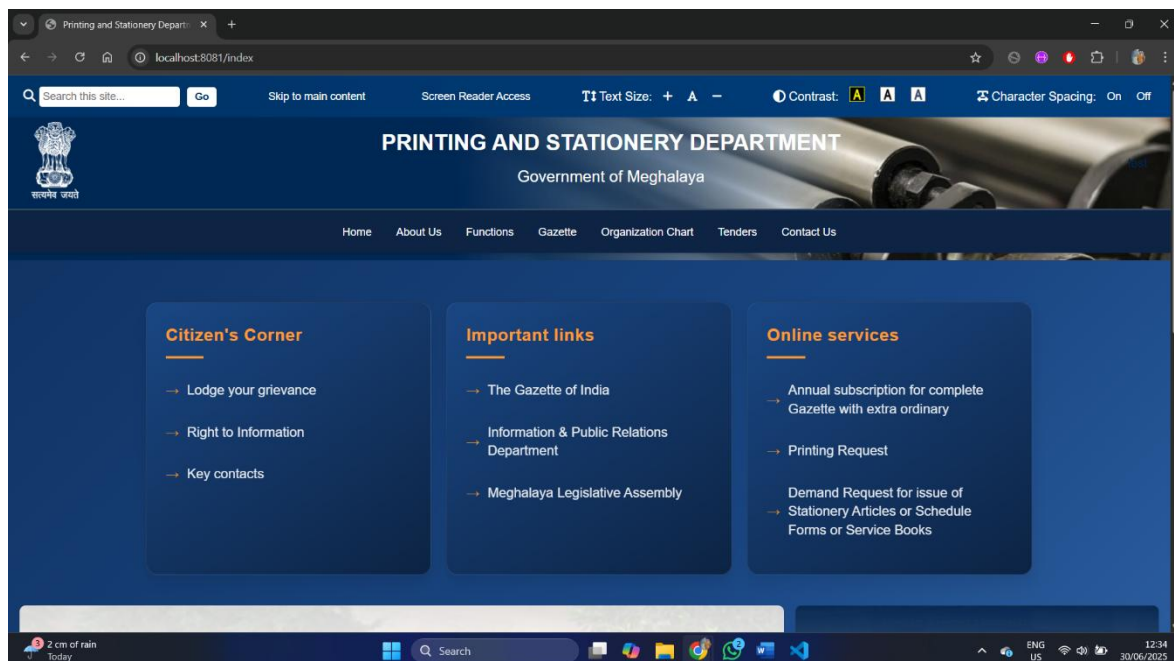
SCREEN SHOTS

Schema Diagram:

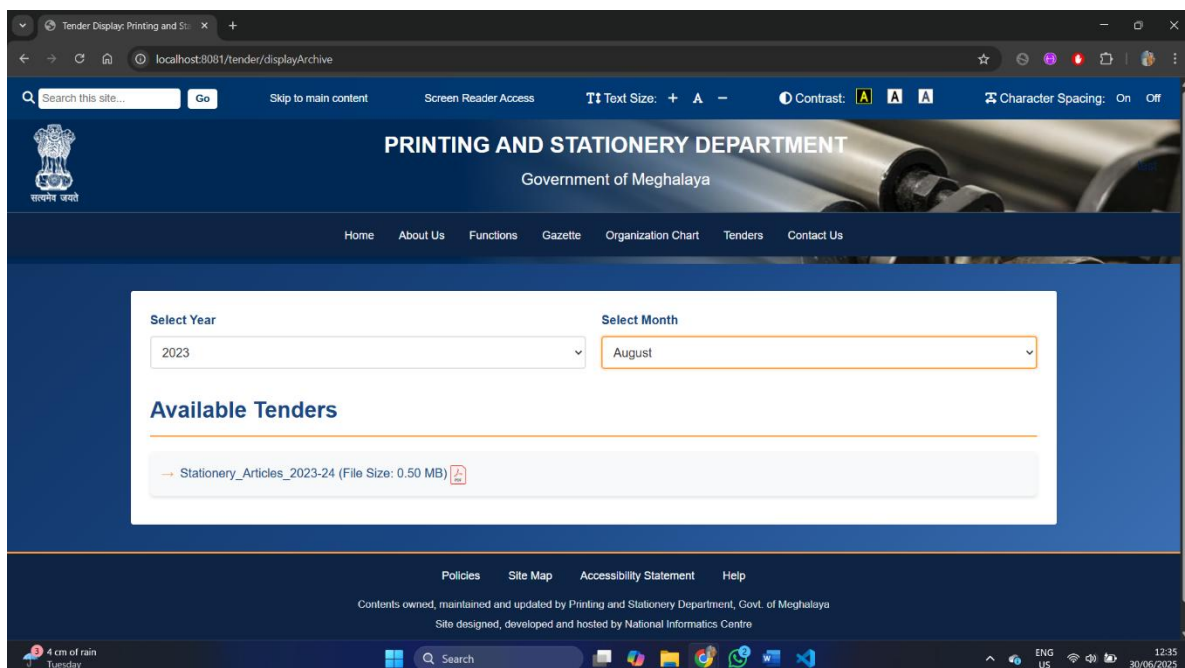
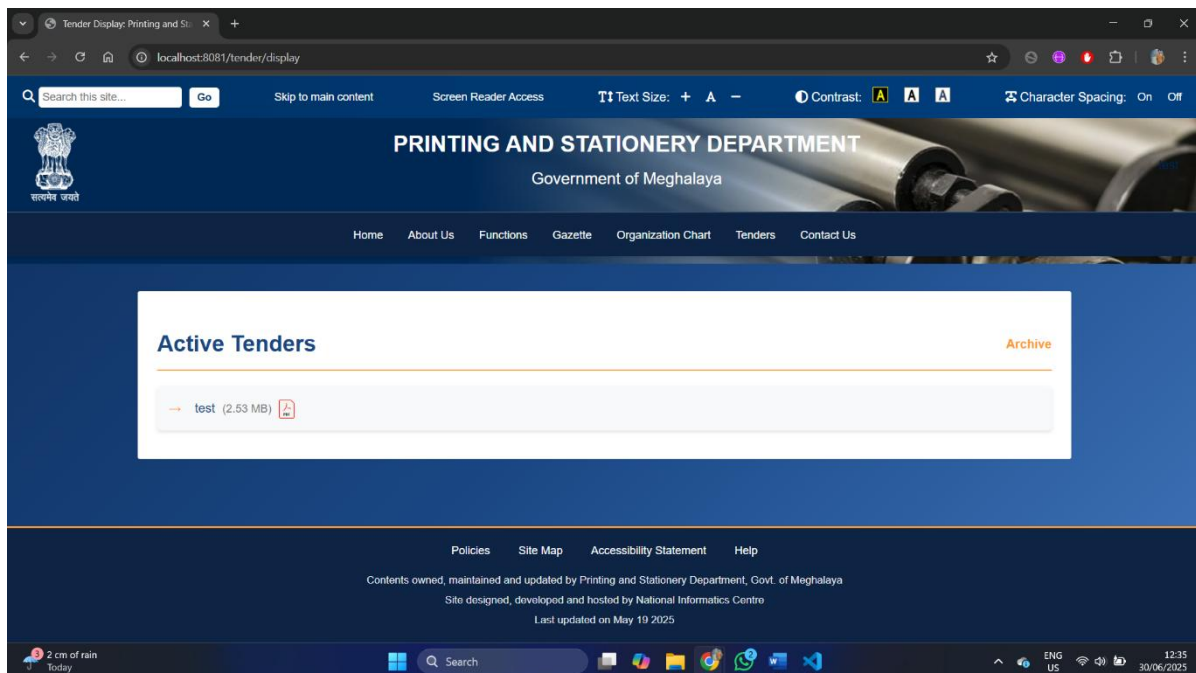


User page Screenshoot:

Home page



Tender view Page



Gazette view page

Gazette Display: Printing and Stationery Department

localhost:8081/gazette/display

Search this site... Go

Skip to main content Screen Reader Access Text Size: + A - Contrast: A A A Character Spacing: On Off

PRINTING AND STATIONERY DEPARTMENT
Government of Meghalaya

Home About Us Functions Gazette Organization Chart Tenders Contact Us

Select Year: 2025 Select Month: January Select Date: 14

Available Gazettes

→ Extraordinary - (File Size: 0.50 MB)

Policies Site Map Accessibility Statement Help

Contents owned, maintained and updated by Printing and Stationery Department, Govt. of Meghalaya
Site designed, developed and hosted by National Informatics Centre

2 cm of rain Today 12:34 30/06/2025

Login Page

Login: Printing and Stationery Department

localhost:8081/login

PRINTING AND STATIONERY DEPARTMENT
Government of Meghalaya

Login

Username

Password

NZpKTm

Enter CAPTCHA

Login

Site designed, developed and hosted by National Informatics Centre

4 cm of rain Tuesday 12:36 30/06/2025

Creator Dashboard

The screenshot shows a web browser window with the URL `localhost:8081/creator`. The page has a dark blue header with the text "PRINTING AND STATIONERY DEPARTMENT" and a user profile "Creator_user". A sidebar on the left contains links: "Upload Gazette", "Gazette Dashboard", "Upload Tender", "Tender Dashboard", "Gazette Submission History", "Tender Submission History", and "Logout". The main content area displays a form titled "Upload Gazette". The form includes a "Select Date" field with a date picker showing "dd/mm/yyyy", a "Select Gazette Part" dropdown menu with "Select Part" selected, an "Upload PDF" section with a "No file selected" button and a "Browse" button, and a "Submit" button at the bottom. The footer of the page reads "© Printing and stationery department meghalaya. All rights reserved." The Windows taskbar at the bottom shows the date as Tuesday, 30/06/2025, and the time as 12:36.

Creator: Printing and Stationery

localhost:8081/creator

PRINTING AND STATIONERY DEPARTMENT

Creator_user

Upload Gazette

Gazette Dashboard

Upload Tender

Tender Dashboard

Gazette Submission History

Tender Submission History

Logout

Upload Gazette

Select Date

dd/mm/yyyy

Select Gazette Part

Select Part

Upload PDF

No file selected

Browse

Submit

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4 cm of rain Tuesday

Search

ENG US

12:36 30/06/2025

The screenshot shows a web browser window with the URL `localhost:8081/creator_tender`. The page has a dark blue header with the text "PRINTING AND STATIONERY DEPARTMENT" and a user profile "Creator_user". A sidebar on the left contains links: "Upload Gazette", "Gazette Dashboard", "Upload Tender", "Tender Dashboard", "Gazette Submission History", "Tender Submission History", and "Logout". The main content area displays a form titled "Upload Tender Details". The form includes a "Tender Title" field with a placeholder "Enter tender title", a "Reference Number" field with a placeholder "Enter reference number", "Announcement Date" and "Last Date for Submission" fields with date pickers showing "dd/mm/yyyy", an "Opening Date" field with a date picker showing "dd/mm/yyyy", an "Upload PDF" section with a "No file selected" button and a "Browse" button, and a "Keywords" field with a placeholder "Enter keywords (comma-separated)". A "Submit" button is at the bottom. The footer of the page reads "© Printing and stationery department meghalaya. All rights reserved." The Windows taskbar at the bottom shows the date as Tuesday, 30/06/2025, and the time as 12:36.

Creator: Printing and Stationery

localhost:8081/creator_tender

PRINTING AND STATIONERY DEPARTMENT

Creator_user

Upload Gazette

Gazette Dashboard

Upload Tender

Tender Dashboard

Gazette Submission History

Tender Submission History

Logout

Upload Tender Details

Tender Title *

Enter tender title

Reference Number *

Enter reference number

Announcement Date *

dd/mm/yyyy

Last Date for Submission *

dd/mm/yyyy

Opening Date *

dd/mm/yyyy

Upload PDF *

No file selected

Browse

Keywords

Enter keywords (comma-separated)

Submit

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localhost:8081/creator

4 cm of rain Tuesday

Search

ENG US

12:36 30/06/2025

Creator: Printing and Stationery

localhost:8081/creator/display

Creator_user

PRINTING AND STATIONERY DEPARTMENT

Upload Gazette

Gazette Dashboard

Upload Tender

Tender Dashboard

Gazette Submission History

Tender Submission History

Logout

Gazette Dashboard

Search by Title Date or Part

Total Tenders: 2

SL NO	TITLE	DATE	PART	ACTIONS
1	2025-06-05-IX.pdf	2025-06-05	IX	More
2	2025-06-26-VIII.pdf	2025-06-26	VIII	More

Edit

Delete

Send

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4 cm of rain
Tuesday

Search

ENG
US

12:37
30/06/2025

Creator: Printing and Stationery

localhost:8081/creator/tender_display

Creator_user

PRINTING AND STATIONERY DEPARTMENT

Upload Gazette

Gazette Dashboard

Upload Tender

Tender Dashboard

Gazette Submission History

Tender Submission History

Logout

Tender Dashboard

Search by Title Reference no or Opening Date

Total Tenders: 1

SL NO	TITLE	REFERENCE NO	ANNOUNCEMENT DATE	LAST DATE	OPENNING DATE	KEYWORDS	ACTIONS
1	Tender title 2	reference number 2	2025-06-30	2025-07-02	2025-07-05	software product, app , web, develop, service,	More

Edit

Delete

Send

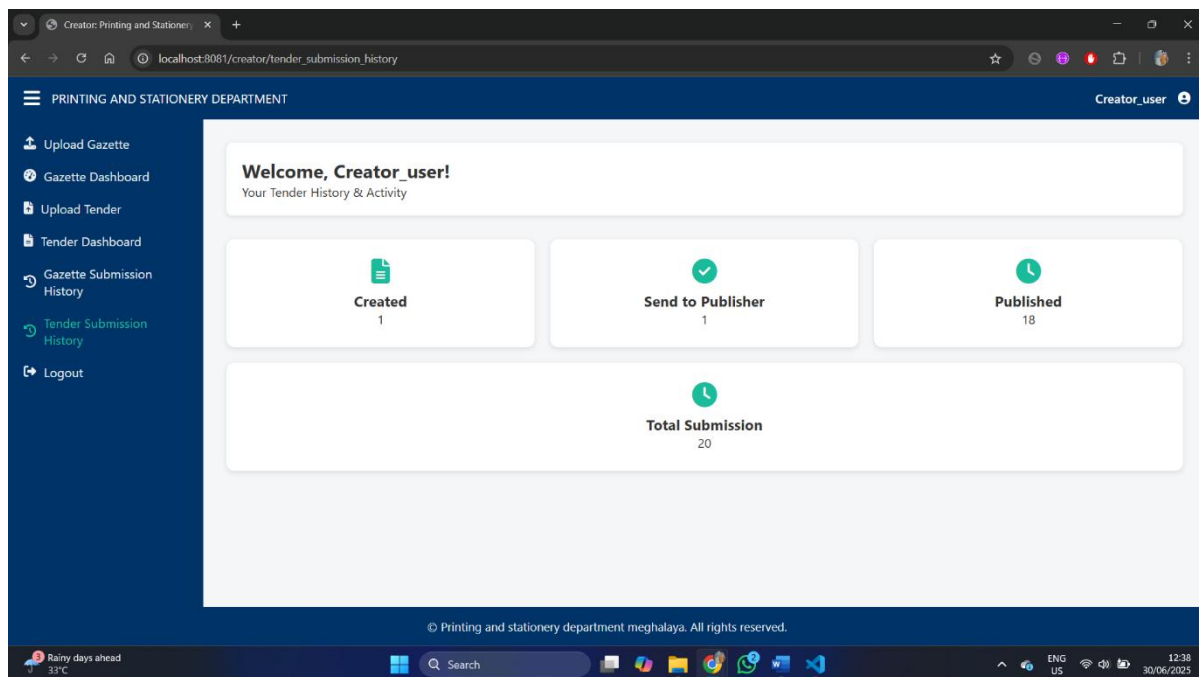
© Printing and stationery department meghalaya. All rights reserved.

Rainy days ahead
33°C

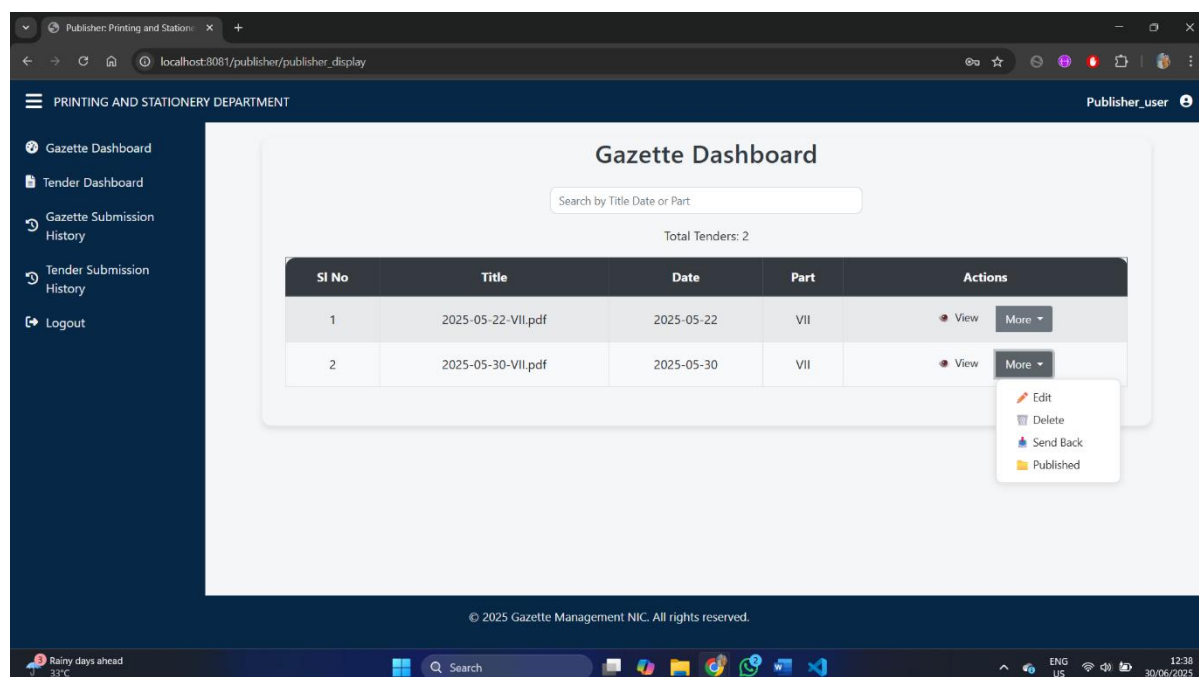
Search

ENG
US

12:38
30/06/2025



Publisher Dashboard



Printer's Printing and Stationery

Gazette Dashboard

Tender Dashboard

Gazette Submission History

Tender Submission History

Logout

Publisher_user

Tender Dashboard

Search by Title Reference no or Opening date

Total Tenders: 1

Sl No	Title	Reference No	Announcement Date	Last Date	Opening Date	Keywords	Actions
1	test	reference number	2025-05-22	2025-05-26	2025-06-01		View More Edit Delete Send Back Published

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Printer's Printing and Stationery

Gazette Dashboard

Tender Dashboard

Gazette Submission History

Tender Submission History

Logout

Publisher_user

Welcome, Publisher_user!

Your Tender History & Activity

Pending Approved

1

Approved

18

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Admin Dashboard

PRINTING AND STATIONERY DEPARTMENT

Gazette Dashboard

Tender Dashboard

Creator List

Publisher List

Upload PDF

PDF Dashboard

Photo Gallery

Contact Us

Logout

Admin_user

Gazett Dashboard

Search by Title Date or Part

Total Tenders: 14

Sl No	Title	Date	Part	Actions
1	2025-05-23-VII.pdf	2025-05-23	VII	More
2	2025-01-02-I.pdf	2025-01-02	I	More
3	2025-01-02-IIA.pdf	2025-01-02	IIA	More
4	2025-01-02-IX.pdf	2025-01-02	IX	More
5	2025-01-09-I.pdf	2025-01-09	I	DeleteView

PreviousNext

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PRINTING AND STATIONERY DEPARTMENT

Gazette Dashboard

Tender Dashboard

Creator List

Publisher List

Upload PDF

PDF Dashboard

Photo Gallery

Contact Us

Logout

Admin_user

Tender Dashboard

Search by Title Reference no or Opening date

Total Tenders: 18

Sl No	Title	Reference No	Announcement Date	Last Date	Opening Date	Keywords	Actions
1	TENDER_NOTICE_18oct2018	DIR/PTG/(s) 8/2018-2019/2	2018-10-17	2018-11-20	2018-11-20		More
2	Tender_printing_Materials_2022-2023	DIR/PTG/(S) 9/2022-2023/4	2022-09-06	2022-10-11	2022-10-11		More
3	Tender_printing_papers_sep_2018	DIR/PTG/(S) 6/2018-2019/1	2018-09-24	2018-10-24	2018-10-24		More
4	Tender_stationery_2013	DIR/PTG/(S)12/2013 – 2014/2	2013-10-09	2013-11-11	2013-11-11		More
5	Tender_Stationery_Articles_2022-2023	DIR/STY/2022-23/4/2	2022-08-29	2022-09-29	2022-09-29		DeleteView

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pg. 25

Admin: Printing and Stationery

localhost:8081/admin/admin_creator_list

PRINTING AND STATIONERY DEPARTMENT

Admin_user

Gazette Dashboard
Tender Dashboard
Creator List
Publisher List
Upload PDF
PDF Dashboard
Photo Gallery
Contact Us
Logout

Creator Dashboard

+ Add Creator

Sl No	Name	Joined Date	Actions
1	Creator_user	2025-04-18	More ▾
2	Creator_user2	2025-04-19	More ▾
3	Testing_user	2025-04-29	More ▾
4	Test_creator	2025-05-21	More ▾
5	Creator_user777	2025-06-18	More ▾

Edit
Delete

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NIFTY -0.25%

Search

ENG US 12:39 30/06/2025

Admin: Printing and Stationery

localhost:8081/admin/admin_publisher_list

PRINTING AND STATIONERY DEPARTMENT

Admin_user

Gazette Dashboard
Tender Dashboard
Creator List
Publisher List
Upload PDF
PDF Dashboard
Photo Gallery
Contact Us
Logout

Publisher Dashboard

+ Add Publisher

Add New Publisher

Username
Enter User Name

Password
Enter Password

Confirm Password
Confirm Password

Admin Password

Submit

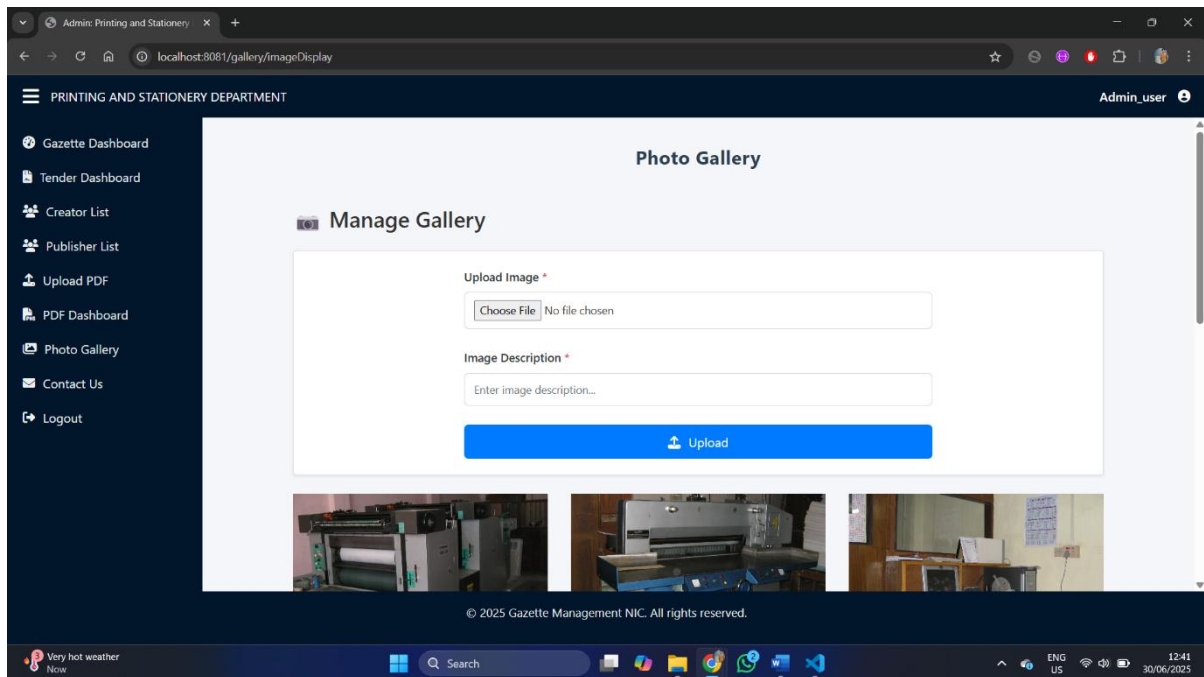
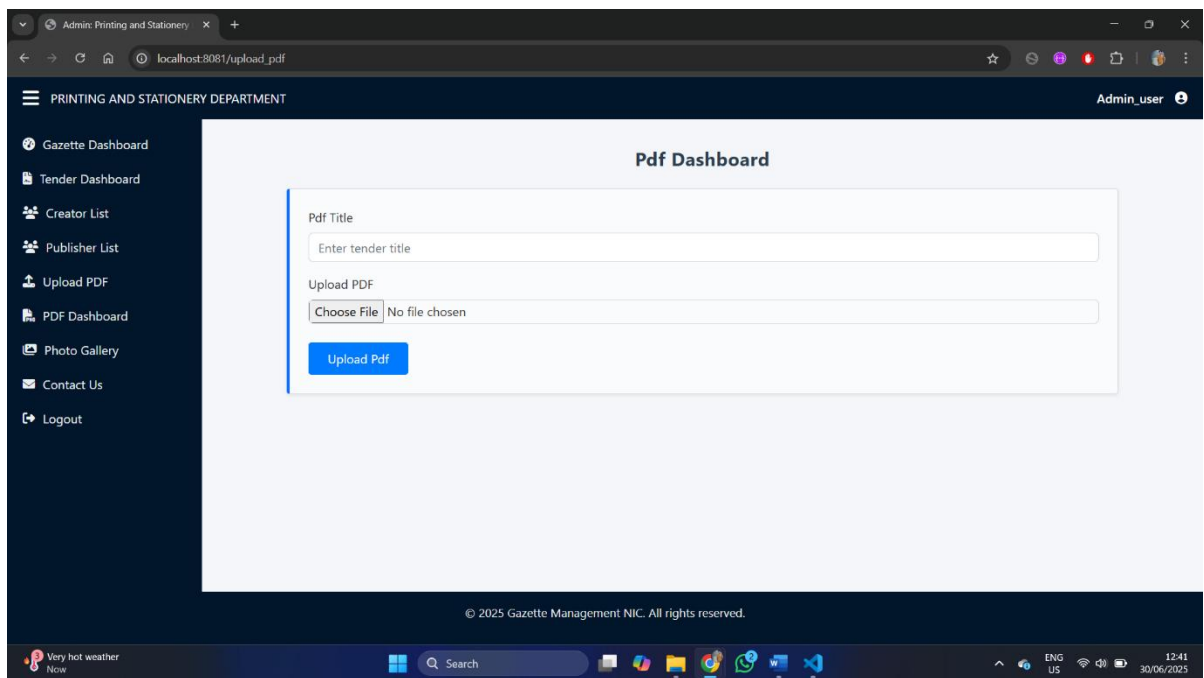
Sl No	Name	Actions
1	Publisher	More ▾

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Hot days ahead 33°C

Search

ENG US 12:40 30/06/2025



SECURITY MEASURES

Ensuring the security of the **Gazette and Tender Management System** is crucial to protect sensitive government documents, safeguard user data, and maintain trust in the digital operations of the Printing and Stationery Department, Government of Meghalaya. The following security measures have been implemented across the application:

1. Encryption:

- **Description:** Sensitive information, including user credentials, uploaded gazette and tender documents, and personal data, is encrypted using strong cryptographic algorithms to prevent unauthorized access during storage and transmission.
- **Implementation:**
 - Use robust encryption libraries in the backend (Spring Boot) to implement encryption for data at rest and in transit.
 - Apply BCrypt hashing for password storage to protect user credentials.
 - Ensure secure document storage to prevent unauthorized file access.

2. Authentication and Authorization:

- **Description:** Strong authentication and authorization mechanisms are employed to verify user identities and ensure that only authorized personnel can access specific system features and documents based on their roles (e.g., Admin, Creator, Publisher).
- **Implementation:**
 - Leverage Spring Security for secure login processes and user session management.
 - Implement role-based access control (RBAC) to define granular permissions for different user types, ensuring that sensitive functions like gazette publishing or tender approvals are restricted to authorized roles.

3. Secure Communication:

- **Description:** All communication between the web portal and backend servers is secured to protect data from eavesdropping and tampering.
- **Implementation:**
 - Enforce HTTPS (SSL/TLS) for all data exchanges between client browsers and the server.
 - Install and maintain SSL/TLS certificates to ensure the authenticity of the server and encrypt transmitted data.

4. Input Validation:

- **Description:** User inputs are validated rigorously to prevent attacks like SQL injection, cross-site scripting (XSS), and other injection vulnerabilities.
- **Implementation:**
 - Implement comprehensive server-side input validation and sanitization for all user-submitted data.
 - Use parameterized queries and prepared statements for all database operations to eliminate SQL injection risks.
 - Validate and restrict file types and sizes during document uploads to avoid malicious content injection.

5. Security Headers:

- **Description:** HTTP security headers are configured to reduce the attack surface of the web application and defend against common web threats.
- **Implementation:**
 - Apply Content Security Policy (CSP) headers to restrict the loading of external scripts and mitigate XSS attacks.
 - Use X-Content-Type-Options to prevent MIME type sniffing.
 - Employ X-Frame-Options to protect the system against clickjacking attacks.

6. Security Testing:

- **Description:** Regular security assessments are conducted to identify and mitigate vulnerabilities proactively, ensuring the system remains secure as it evolves.
- **Implementation:**
 - Use automated vulnerability scanning tools to detect weaknesses in application code and configurations.
 - Conduct manual penetration testing to simulate real-world attack scenarios and identify potential threats.
 - Perform regular code reviews focused on secure coding practices to catch vulnerabilities early in development.

By diligently applying these security measures, the **Gazette and Tender Management System** ensures the confidentiality, integrity, and availability of sensitive government data, providing a secure and trusted platform for managing gazettes and tenders under the Printing and Stationery Department, Government of Meghalaya.

FUTURE SCOPE OF THE SYSTEM

The Gazette and Tender Management System for the Printing and Stationery Department, Government of Meghalaya, has been developed as a robust digital solution to manage gazette publications, tender notices, and related workflows efficiently. However, there are several potential areas for future enhancements and innovations to further improve its capabilities, security, usability, and integration with wider government initiatives:

1. Advanced Analytics and Reporting:

- Integrate powerful analytics dashboards to provide insights into gazette publications, tender submissions, and user engagement. Enable the generation of customized reports for administrative review, helping in decision-making and trend analysis related to publication frequencies, departmental activities, and document access patterns.

2. Digital Signatures and E-Authentication:

- Incorporate digital signature support to authenticate gazette publications and tender documents officially. Implement secure electronic signing for authorized users to enhance legal validity, eliminate manual signing processes, and ensure document integrity.

3. Integration with Government e-Procurement Portals:

- Integrate the system with state and national e-Procurement portals to seamlessly publish tenders, receive bids, and synchronize data. This will promote unified tender management and reduce redundancy between departmental systems and central government platforms.

4. Mobile Application Development:

- Develop a dedicated mobile app to provide on-the-go access for users to view gazette notifications, download documents, receive tender alerts, and submit bids. Mobile accessibility would enhance user convenience and broaden reach, especially for stakeholders in remote areas.

5. Role-Based Document Workflow Management:

- Implement advanced workflow management features for gazette publication and tender processes. Enable tracking of document statuses, approvals, and version histories to ensure accountability and transparency in administrative workflows.

6. Multilingual Support:

- Extend language support to include Khasi, Garo, and other regional languages of Meghalaya. Multilingual content would ensure greater inclusivity and accessibility for local citizens, fostering better communication and engagement.

7. Public Feedback and Interaction Platform:

- Develop features allowing citizens and stakeholders to provide feedback or comments on published gazettes or tenders. This would promote public participation and transparency in governance.

8. Document Archival and Retrieval System:

- Enhance archival capabilities with intelligent search and retrieval mechanisms to manage historical gazettes and tenders efficiently. Enable fast retrieval using metadata, keywords, and filters for improved user experience and administrative convenience.

9. Artificial Intelligence for Smart Document Classification:

- Utilize AI and machine learning algorithms to classify gazette notifications and tender documents automatically into predefined categories. This would reduce manual categorization efforts and improve content discoverability.

10. Enhanced Security Features:

Continuously upgrade security measures to protect sensitive documents. Future security enhancements may include:

- Biometric authentication for high-privilege roles
- Blockchain integration for tamper-proof publishing history
- Real-time threat detection and automated incident responses

By adopting these future enhancements, the **Gazette and Tender Management System** can evolve into a comprehensive, citizen-centric platform that modernizes document management, promotes transparency, and fosters greater administrative efficiency for the Printing and Stationery Department, Government of Meghalaya.

RESULTS AND CONCLUSION

Results:

1. **Improved Document Management and Accessibility:** The development and implementation of the Gazette and Tender Management System have significantly enhanced the management and accessibility of official documents. Users can now easily browse, download, and access gazette notifications and tender documents online, reducing reliance on physical paperwork.
2. **Streamlined Publication Process:** The system has streamlined the entire publication workflow for gazettes and tenders, enabling efficient uploading, updating, and dissemination of official documents by authorized personnel. This has reduced delays and improved the speed of information dissemination.
3. **Enhanced Transparency and Accountability:** By digitizing the publication process, the system has increased transparency in the release of government notifications and tenders. Stakeholders and citizens can access information directly from a centralized platform, reducing ambiguity and ensuring accountability in government operations.
4. **Role-Based Access Control:** The system's implementation of role-based access control has ensured secure management of content and administrative actions. Different user roles (Admin, Creator, Publisher) help maintain data integrity and prevent unauthorized modifications to sensitive information.
5. **Robust Security Measures:** The integration of modern security measures—including authentication, encryption, and secure communication protocols—has safeguarded user data, document integrity, and the privacy of confidential government information.
6. **Operational Efficiency:** The system has reduced manual tasks, paperwork, and physical storage requirements, enabling more efficient administrative operations within the Printing and Stationery Department. This efficiency contributes to cost savings and better resource utilization.

Conclusion:

In conclusion, the **Gazette and Tender Management System** developed for the Printing and Stationery Department, Government of Meghalaya, has proven to be a significant step toward modernizing government workflows. By transitioning to a digital platform, the department has successfully improved document management, enhanced transparency, and provided citizens and stakeholders with convenient access to vital government publications.

The system's role-based architecture, security features, and user-friendly interface have contributed to its effectiveness and reliability. It has not only reduced manual efforts and operational bottlenecks but also fostered public trust by making official information more accessible and timelier.

Looking ahead, continuous improvement and the integration of advanced features such as analytics dashboards, multilingual support, mobile accessibility, and integration with other government systems can further enhance the platform's capabilities and outreach. Collaboration with other departments and stakeholders will be crucial to ensure the system remains scalable, user-centric, and compliant with evolving regulations and technological standards.

Overall, the **Gazette and Tender Management System** represents a significant leap toward e-governance in Meghalaya, contributing to greater efficiency, transparency, and public engagement in government processes. It lays a solid foundation for future digital initiatives aimed at strengthening governance and public service delivery across the state.

Bibliography & References

- <https://www.w3schools.com>
- <https://stackoverflow.com>
- <https://www.tutorialspoint.com>
- <https://www.youtube.com>
- <https://chatgpt.com>
- <https://www.blackbox.ai/>
- Java Programming by K. Rajkumar